Business Requirements Document

DailyBasket Data Warehouse - Phase 1

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1. Purpose

This document lists what DailyBasket management needs from the data warehouse. It is the base for the dimensional design in Phase 1 and the build in Phase 2.

2. Business Background

DailyBasket sells groceries through three channels (In-Store, Website, Mobile App) in several cities. Management wants to grow revenue, keep good customers, avoid stock problems and use promotions wisely.

3. Stakeholders

Stakeholder	Role	What they need
General Manager	Overall decisions	Revenue and profit trends, regional growth
Sales Manager	Sales and channels	Top products, weak regions, channel performance
Marketing Manager	Campaigns and promotions	Which promotions work, customer segments
Inventory Manager	Stock and purchasing	Low stock, over-stock, declining demand
Customer Manager	Customer care and loyalty	Lifetime value, repeat purchase behavior

4. Business Objectives

- Increase sales revenue by finding top products, channels and regions.
- Increase customer value by knowing the best customers and segments.
- Reduce stock-outs and over-stock by watching inventory.
- Spend promotion budget on offers that really raise sales.

5. Scope

In scope: Sales Transactions, Inventory Snapshots, Promotion Events. Dimensions: Date, Customer, Product, Region, Channel, Promotion.

Out of scope: real-time data, payments, supplier data, employee data.

6. Source Systems

Source	Data	Format	Refresh
Order system	Orders and order lines	orders.csv	Daily
Customer master	Customer details and segment	customers.csv	Daily
Product catalog	Products, category, brand, cost, price	products.csv	Weekly
Inventory system	Stock levels per product and region	inventory.csv	Daily
Promotion system	Promotions and the products in them	promotions.csv	As needed

Sample data will be generated with Python/Pandas in Phase 2 so the work can be repeated by anyone.

7. Functional Requirements

ID	Requirement	Priority
FR1	The warehouse must show net revenue and profit by day, month, quarter and year.	High
FR2	The warehouse must show sales by product, subcategory and category.	High
FR3	The warehouse must show sales by region (city, province, country).	High
FR4	The warehouse must show sales by channel.	High
FR5	The warehouse must show customer lifetime value and purchase frequency by segment.	High
FR6	The warehouse must keep a daily stock snapshot per product and region.	High
FR7	The warehouse must show which products and days were covered by each promotion.	Medium
FR8	The warehouse must let users compare sales with and without promotions.	Medium
FR9	Every business question in the Business Questions document must be answerable with SQL.	High

8. Key Performance Indicators (KPIs)

KPI	Meaning
Net Revenue	Gross amount minus discount
Profit	Net revenue minus cost
Quantity Sold	Number of units sold
Average Order Value	Net revenue divided by number of orders
Customer Lifetime Value	Total net revenue from one customer
Stock on Hand / Stock Value	Units in stock and their cost value
Promotion Uplift	Sales during promotion compared with normal days

9. Non-Functional Requirements

- **History:** keep about 2 years of data.
- **Data quality:** no duplicate keys, no null keys, valid dates.
- **Repeatable:** all SQL and scripts saved in GitHub.
- **Cost:** use free tools only.
- **Performance:** star schema so common queries stay simple and fast.

10. Assumptions and Constraints

- Currency is Pakistani Rupees (PKR).
- Each order line has one product and belongs to one order.
- A product can be in only one promotion at a time on a given day.
- Data is loaded daily (batch), not in real time.

11. Success Criteria

- All business questions can be answered from the warehouse.

- The star schema has clear grain, keys, dimensions and measures.
- The model is approved before implementation starts in Phase 2.